Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/16/2018
PI Name:	Gilbert, Rachel Ph.D.		
Project Title:	Simulating and Measuring Spaceflight-Induced Changes in a Host-Pathogen System		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	04/17/2017	**End Date:**	04/16/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
Contact Email:	npphelp@usra.edu		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (MENTOR/ NASA Ames Research Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			
	NASA Postdoctoral Program Fellow While recent advances in technology have made it possible for humans to achieve long---term space exploration, there is still relatively little known about the effects of spaceflight conditions on human physiology. While it has been shown that decades of astronauts and cosmonauts suffer from immune disorders both during and after spaceflight, the underlying causes are still poorly understood, due in part to the fact that there are so many variables to consider when investigating the human immune system in a complex environment. Invertebrates have become popular models for studying human disease because they are cheap, highly amenable to experimental manipulation, and have immune systems with a high genetic similarity to humans. Fruit flies (Drosophila melanogaster) have been shown to experience a dramatic shift in immune gene expression following spaceflight, but are still able to fight off infections when exposed to		

Task Description:	bacteria. However, the common bacterial pathogen Serratia marcescens was shown to become more lethal to fruit flies after being cultured in space, suggesting that not only do we need to consider host changes in susceptibility, but also changes in the pathogen itself after spaceflight conditions. Being able to simulate spaceflight conditions in a controlled environment on the ground gives us the ability to not only evaluate the effects of microgravity on the host immune system, but also how the microorganisms that cause immune disorders are being affected by these drastic environmental shifts. In this study, I propose to use a simulated microgravity environment to examine the genetic changes associated with increased S. marcescens virulence in order to understand how microgravity is affecting this pathogen. After determining which genes are being expressed differentially, I will then use available molecular tools to manipulate these spaceflight--related genes one at a time. Infecting fruit flies with the modified strains and measuring the host immune response to these modified bacteria will allow us to better understand how spaceflight-induced changes to pathogens are associated with increased host lethality. This study will provide us with more directed approaches to studying the effects of spaceflight on human beings, with the ultimate goal of being able to prevent human immune dysfunction in future space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2017.
Bibliography Type:	Description: (Last Updated: 10/08/2023)
Articles in Peer-reviewed Journals	Iyer J, Mhatre SD, Gilbert R, Bhattacharya S. "Multi-system responses to altered gravity and spaceflight: Insights from Drosophila melanogaster." Neurosci Biobehav Rev. 2022 Nov;142:104880. Review. https://doi.org/10.1016/j.neubiorev.2022.104880 ; PMID: 36126744 , Nov-2022
Articles in Peer-reviewed Journals	Gilbert R, Tanenbaum N, Bhattacharya S. "Asparagine biosynthesis as a mechanism of increased host lethality induced by Serratia marcescens in simulated microgravity environments." Heliyon. 2022 May 1;8(5):e09379. https://doi.org/10.1016/j.heliyon.2022.e09379 ; PMID: 35592661; PMCID: PMC9111996 , May-2022